

Ratio: sharing and missing amounts

Worked demo • Follow each step down your column.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A and B share £120 in the ratio 2:3. How much does B receive?	Red and blue paint are mixed in the ratio 3:5. There are 18 litres of red paint. How many litres of blue paint are needed?	Red and blue counters are in the ratio 3:7. There are 28 more blue counters than red counters. How many counters are there altogether?
Keywords & Calculation	“share £120”: the total is known. $2 + 3 = 5$ parts. $£120 \div 5$, then $\times 3$.	“18 litres of red”: 3 parts are known. $18 \div 3$, then $\times 5$.	“28 more blue”: the difference is known. $7 - 3 = 4$ parts. $28 \div 4$, then $\times (3 + 7)$.
Written Method	$£120 \div 5 = £24$ per part. $£24 \times 3 = £72$. A B £120 altogether 1 part = £24	$18 \div 3 = 6$ litres per part. $6 \times 5 = 30$ litres. Red Blue Red: 18 litres 1 part = 6 litres	$28 \div 4 = 7$ counters per part. $3 + 7 = 10$ parts. $7 \times 10 = 70$ counters. Red Blue 4 extra parts = 28 counters 1 part = 7 counters
Ballpark Check & Math	B gets more than half of £120, so more than £60. £72 fits. Check both shares: $£48 + £72 = £120$.	Blue is between one and two times the red amount, so between 18 and 36 litres. 30 litres fits.	The total represents 10 parts and the difference 4. The total should be between 2 and 3 times 28: between 56 and 84. 70 fits.
Final Answer	B receives £72.	30 litres of blue paint.	70 counters altogether.

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Step-by-step practice 1 • Complete each step in your booklet or webapp.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	An 84 ml paint mix contains red and white paint in the ratio 3:4. How much red paint is in the mix?	Notebooks and folders are in the ratio 4:7. There are 35 folders. How many notebooks are there?	Adults and children are in the ratio 2:5. There are 24 more children than adults. How many people are there altogether?
Keywords & Calculation	Identify the keywords. Write down the calculation(s) you need to do.	Identify the keywords. Write down the calculation(s) you need to do.	Identify the keywords. Write down the calculation(s) you need to do.
Written Method	Set up and use the written method for your calculation(s).	Set up and use the written method for your calculation(s).	Set up and use the written method for your calculation(s).
Ballpark Check & Math	Check your answer using rough ballpark calculations. Does it make sense?	Check your answer using rough ballpark calculations. Does it make sense?	Check your answer using rough ballpark calculations. Does it make sense?
Final Answer	Write your final answer to the question. Include the correct units.	Write your final answer to the question. Include the correct units.	Write your final answer to the question. Include the correct units.

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Step-by-step practice 2 • Complete each step in your booklet or webapp.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A 96 cm ribbon is cut into pieces A and B in the ratio 5:7. Find the length of B.	Juice and water are mixed in the ratio 2:7. There are 18 litres of juice. How many litres of water are needed?	Two lengths of ribbon, A and B, are in the ratio 5:8. B is 21 cm longer than A. What is their total length?
Keywords & Calculation	Identify the keywords. Write down the calculation(s) you need to do.	Identify the keywords. Write down the calculation(s) you need to do.	Identify the keywords. Write down the calculation(s) you need to do.
Written Method	Set up and use the written method for your calculation(s).	Set up and use the written method for your calculation(s).	Set up and use the written method for your calculation(s).
Ballpark Check & Math	Check your answer using rough ballpark calculations. Does it make sense?	Check your answer using rough ballpark calculations. Does it make sense?	Check your answer using rough ballpark calculations. Does it make sense?
Final Answer	Write your final answer to the question. Include the correct units.	Write your final answer to the question. Include the correct units.	Write your final answer to the question. Include the correct units.

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Independent practice 1 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1 Sam and Jo share £60 in the ratio 2:3. How much does Jo receive?	1 Flour and sugar are mixed in the ratio 3:2. A recipe uses 180 g of flour. What is the total mass of the mixture?	1 Two ropes have lengths in the ratio 1:3. The longer rope is 12 m longer than the shorter. Find their total length.
2 A drink contains 600 ml of juice and water in the ratio 1:4. How much juice is in the drink?	2 There are 64 adult and child tickets in the ratio 5:3. How many tickets are for children?	2 Red and blue counters are in the ratio 1:5. There are 24 fewer red than blue. Find the number of each colour.
3 A 96 cm ribbon is cut into two pieces in the ratio 5:7. Find the length of each piece.	3 Ropes A and B have lengths in the ratio 4:7. B is 15 m longer than A. Find the length of A.	3 Ava and Leo share £100 in the ratio 7:3. How much more does Ava receive than Leo?
4 A bag has 48 red and blue beads in the ratio 3:5. How many blue beads are in the bag?	4 Mia and Ben receive money in the ratio 2:5. Mia receives £18. How much do they receive altogether?	4 Flour and sugar have masses in the ratio 5:4. There is 300 g of sugar. Find the total mass.
5 Flour and sugar are mixed in the ratio 5:2. A recipe uses 100 g of sugar. Find the mass of flour.	5 Juice and water are mixed in the ratio 2:7. There are 63 litres of water. How much juice is needed?	5 A 1,400 ml drink has water and juice in the ratio 6:1. Find the volume of each ingredient.
6 Practice and rest times are in the ratio 2:1. Practice lasts 36 minutes. How long is the rest?	6 Travel and sightseeing take 55 minutes in total, in the ratio 3:8. How long is spent sightseeing?	6 Practice and rest times are in the ratio 6:5. Practice lasts 54 minutes. Find the total session time.

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Independent practice 2 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1 A 560 g snack mix contains raisins and nuts in the ratio 1:7. Find the mass of raisins.	1 A walk has uphill and level sections in the length ratio 1:8. The uphill section is 3 km. Find the total distance.	1 Flour and butter have masses in the ratio 9:2. There is 280 g less butter than flour. Find the total mass.
2 A 117 m fence is split between gardens A and B in the ratio 7:6. Find the length for each garden.	2 A 490 ml paint mix contains red and white paint in the ratio 3:4. Find the volume of red paint.	2 Tanks A and B hold water in the ratio 3:8. Tank A holds 15 litres. How much more water is in B than A?
3 A 45-minute lesson has instruction and practice in the ratio 1:2. How long is spent practising?	3 Eve and Max receive money in the ratio 8:5. Eve gets £21 more than Max. Find Max's share.	3 Reading and writing take 78 minutes in total, in the ratio 4:9. Find both times.
4 Kim and Ali receive money in the ratio 7:5. Kim gets £56. How much does Ali receive?	4 Study and break times are in the ratio 5:1. Breaks total 12 minutes. Find the total study and break time.	4 Two pieces of wood have lengths in the ratio 5:9. The longer is 28 cm longer. Find the longer length.
5 A planter has 63 red and white flowers in the ratio 4:5. How many flowers are white?	5 A 300 g mix contains raisins and oats in the ratio 7:8. Find the mass of raisins.	5 Nia and Tom share £208 in the ratio 7:9. How much more does Tom get than Nia?
6 Blue and white paint are mixed in the ratio 1:3. There is 180 ml of white paint. Find the total volume.	6 A farm has sheep and goats in the ratio 9:1. There are 72 sheep. How many animals are there altogether?	6 Adult and child tickets sell in the ratio 9:8. There are 81 adult tickets. Find the total number of tickets.

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Independent practice 3 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1 A club has 80 junior and senior members in the ratio 7:3. How many members are seniors?	1 Study and break times are in the ratio 8:1. Study takes 64 minutes. Find the total time.	1 Two sessions have durations in the ratio 1:10. The second lasts 72 minutes longer. Find the first session's duration.
2 Milk and coffee are mixed in the ratio 4:1. There is 100 ml of coffee. Find the volume of milk.	2 Metal rods A and B have lengths in the ratio 5:8 and total 91 mm. Find the length of B.	2 Kai and May receive money in the ratio 9:5. May gets £60. Find their combined amount.
3 Dan and Pat share £98 in the ratio 1:6. Find each person's share.	3 A recipe uses flour and butter in the ratio 4:3 by mass. There are 90 g of butter. Find the total mass.	3 A farm has 143 sheep and goats in the ratio 10:3. How many more sheep are there than goats?
4 A 165-minute film has action and dialogue in the ratio 5:6. How many minutes are dialogue?	4 Green and yellow beads are in the ratio 7:8. There are 3 fewer green beads. Find the number of yellow beads.	4 Milk and water have volumes in the ratio 10:7. There is 500 ml of milk. How much less water is there than milk?
5 A 800 g fruit mix contains apples and pears in the ratio 7:1. Find the mass of pears.	5 Sam and Jo share £90 in the ratio 9:1. How much does Jo receive?	5 Two paths have lengths in the ratio 4:9. The longer is 35 m longer. Find their combined length.
6 Wire pieces A and B have lengths in the ratio 7:6. A is 84 cm long. Find the length of B.	6 Concentrate and water are mixed in the ratio 2:9. There is 675 ml of water. Find the total volume.	6 A 950 g mix contains oats and raisins in the ratio 10:9. Find the mass of each ingredient.

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Independent practice 4 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1 A 30-minute trip has walking and cycling times in the ratio 1:2. Find the cycling time.	1 Two charities receive money in the ratio 7:9. The second gets £108. How much is donated altogether?	1 Two tanks hold water in the ratio 1:11. The second holds 90 litres more. How much water is in the first?
2 Two buckets contain 20 litres of water in total, in the ratio 1:3. Find the amount in each bucket.	2 A tray has 102 red and yellow plants in the ratio 8:9. How many plants are yellow?	2 Nuts and raisins are mixed in the ratio 11:7 by mass. There are 210 g of raisins. Find the total mass.
3 Ribbon pieces A and B have lengths in the ratio 2:3. A is 24 cm long. Find the length of B.	3 Mixing and baking times are in the ratio 1:5. Baking takes 25 minutes. How long does mixing take?	3 Two ribbons have a total length of 285 cm, in the ratio 8:11. Find the difference in their lengths.
4 Noor and Will receive money in the ratio 7:4. Will gets £48. Find Noor's share.	4 Two sacks have masses in the ratio 7:3. The heavier is 16 kg heavier. Find the lighter sack's mass.	4 Adult and child tickets sell in the ratio 11:2. There are 143 adult tickets. How many more adult tickets are sold?
5 A box has 60 dark and milk chocolates in the ratio 1:4. Find the number of each type.	5 Blue and white paint are mixed in the ratio 2:5. There are 15 litres of white paint. Find the total volume.	5 Leah and Omar receive money in the ratio 3:11. Leah gets £64 less than Omar. How much does Omar get?
6 A 560 g bread mix has white and brown flour in the ratio 4:3. Find the mass of brown flour.	6 Ropes A and B total 56 m, with lengths in the ratio 5:3. Find the length of B.	6 Work and breaks take 90 minutes in total, in the ratio 7:2. How much longer is spent working than on breaks?